Naga Venkata Sai Nadh Vaka

Master's student at OTH Regensburg seeking a Working Student/ Internship Role



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Obertraubling, Germany (Flexible to relocate) LinkedIn GitHub

Engineering Portfolio

PROFILE

Master's student in Electrical and Microsystems Engineering at OTH Regensburg with a strong interest in semiconductor technology, electronics, and embedded systems. Developing hands-on experience in hardware design, laboratory testing, and data analysis through engineering projects.

EDUCATION

03/2026 – Present
Regensburg, Germany

Master of Engineering in Electrical and Microsystems Engineering,
Ostbayerische Technische Hochschule (OTH) Regensburg

- **Relevant Coursework:** Advanced Semiconductor Technology, Electronic Circuits & Systems, Electronic Product Engineering, Advanced Packaging, Applied Optics.

12/2021 – 04/2025
Vijayawada, India

Bachelor of Technology in Electronics and Communication Engineering,
Usha Rama College of Engineering and Technology

- **Relevant Coursework:** Digital IC Design, VLSI Design, Linear Integrated Circuits & Applications, Electronic Devices & Circuits, Electronic Circuit Analysis.
- **Final Grade:** 1.6 - Graduated with Distinction.
- **Qualified GATE 2024** (EC).

EXPERIENCE

11/2024 – 04/2025
Vijayawada, India

Embedded Systems with IoT Applications Intern, *Protective Technologies Pvt. Ltd.*

- **Supported** development and testing of embedded systems using microcontrollers and sensor interfaces.
- **Performed** system validation, troubleshooting, and integration activities for embedded and IoT-based applications.
- **Documented** technical activities and supported engineering workflows in collaboration with the project team.

08/2024 – 04/2025
Vijayawada, India

President - ECRATZ ECE Student Association,
Usha Rama College of Engineering and Technology,

- Elected to represent **700+ engineering students** and coordinated technical events with faculty and student volunteers.

TECHNICAL SKILLS

Electronics & Hardware: Electronic Components, Electronic Circuits, Embedded Systems, Microcontrollers, Sensor Interfaces, PCB Prototyping

Testing & Measurement: Oscilloscopes, Digital Multimeters, DC Power Supplies, Functional Testing, Troubleshooting

Programming & Data: Python, MATLAB, C, Data Analysis, Image Processing

Digital & Semiconductor Design: Verilog, VHDL, SystemVerilog (Basic), RTL Design, Digital IC Design, CMOS Fundamentals, VLSI Fundamentals, Functional Verification

Simulation & Engineering Tools: AMD Vivado, Git, Keil µVision, Multisim, Proteus, Arduino IDE

Documentation & Collaboration: Technical Documentation, Experimental Reporting, Problem Solving, Teamwork

LANGUAGES

ENGLISH — C1 (Fluent)

GERMAN — B1 (Actively Learning)

TELUGU — Native

PROJECTS

05/2026 – Present

AXI4-Lite SPI Flash Controller, *Verilog | AMD Vivado*

- **Designed** an AXI4-Lite register interface and FSM-based SPI controller enabling memory-mapped Flash access using Verilog.
- **Integrated** SPI transaction sequencing, register decoding, and Flash command execution into a synthesizable RTL design.
- **Verified** the design through custom Verilog testbenches and FPGA implementation, achieving **0 setup and hold timing violations** at a **100 MHz** target clock.
- **Optimized** hardware utilization to **193 LUTs** and **206 Flip-Flops** on a Xilinx Virtex-7 FPGA.

11/2024 – 03/2025

Advanced Image Encryption Algorithm Integrating Chaotic Image Encryption and CNN, *MATLAB | Image Processing | Research Project*

- Automated MATLAB pipeline for a 3,000-image dataset; generated PSNR, SSIM, and histogram metrics; eliminated repetitive manual validation.
- **Developed reusable MATLAB modules** for simulation, convolution-based feature extraction, dataset processing, and algorithm validation.

03/2023 – 05/2023

Transformerless AC-to-DC Power Supply, *Electronics | Laboratory Testing*

- Designed and tested a transformerless AC-to-DC power supply prototype for 5V/9V/12V/15V outputs.
- Performed electrical measurements, troubleshooting, and performance evaluation using laboratory equipment.

PUBLICATIONS

03/2025

Advanced Image Encryption Algorithm Integrating Chaotic Image Encryption and CNN, *i-manager's Journal on Electronics Engineering*

- **Published** a peer-reviewed journal article on a hybrid image encryption framework integrating chaotic cryptography and convolutional neural networks.

LEADERSHIP & ACTIVITIES

09/2022 – 04/2025
Vijayawada, India

Class Representative, *Usha Rama College of Engineering and Technology*

- **Coordinated** communication between faculty and a cohort of **60+ students**, ensuring timely dissemination of academic and administrative information.

INTERESTS

Badminton | Hiking & Nature | Electronics & DIY Projects | Photography